Supplementary Methods

EMToolbox uncovers biased co-option of embryonic EMT programs across cancers

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MesenContext: Agent instruction set

MesenContext uses natural language to encode a decision tree and successive rounds of human-agent knowledge transfer, indirectly steering AI agents during their autonomous analyses of Curated Cancer Cell Atlas (3CA) datasets (Tyler *et al.*, 2025). Each section of these methods was a separate specification file in `mesencontext/spec/`, reproduced below with cosmetic updates. Section 1 defines the analysis task, Sections 2–4 audit agent performance on steered and autonomous analyses, and Section 5 logs all runs across 3CA datasets. The documented context was accumulated as the project progressed — each agent received every section completed before its run, including newly added subsections and ledger updates. These specification files were AI-generated with human steering to ensure scientific and methodological accuracy.

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1 Cell-state analysis: Task definition

0. Overview

Re-run a six-step single-cell analysis — hand-built on Puram2017 (HNSCC) — across every in-scope 3CA dataset, one per subdirectory.

Question: Do malignant cells recapitulate embryonic-development programs? Score three modules (Gast, pSom, Crest) on malignant cells, fit a cell-state manifold / trajectory, and use three GO/NO-GO gates to decide whether the expected pattern — a Gast/pSom hub diverging into two arms with cross-repression — reproduces.

Role: This is **not** a fixed pipeline; nearly every parameter is dataset-dependent. Act as the **analyst** — make the same *kinds* of judgment calls made for Puram2017, grounded in the data and biology. The Puram2017 notebooks are the **protocol** (§1.3), not values to copy.

Scope: `CCCA_summary.csv` (§1.4) is authoritative. The on-disk folder set is a superset — folders not in the summary are out of scope; do not analyze them.

1. Biology

Hypothesis: Malignant cells recapitulate **embryonic-development gene programs**; EMT mirrors developmental trajectories and/or bifurcations. Modules (read the `hsapiens` column, drop NaN/dupes):

- **Gast**: Gastrulation (`GSE162534.csv`)
- **pSom**: Posterior somitogenesis (`GSE229103.csv`)
- **Crest**: Neural crest (`rRNAModifications.csv` — filename historical; contents are neural crest)
- **union**: Control pool for `score_genes` (`union.csv`)

Expected pattern (from Puram2017): A **hub** where Gast and pSom co-express **diverges** into a pSom-leaning arm and a Gast-leaning arm. **Crest** is high at the arm tips (terminal readout), enriched on the pSom arm. **Gast↔pSom is the cross-repression axis** — opposite directions out of the hub; mutual exclusivity is the bifurcation signature. Crest is a terminal readout, **not** on the cross-repression axis. Hub module correlation in Puram ≈ -0.49. Module→arm assignment is **data-driven** — read it off the scores, not off subtype names.

Not every dataset bifurcates. In the **non-branching case**, Gast and pSom **partially or fully overlap**, with Crest as an inverse/terminal gradient — the programs are present but not resolved into two competing arms, though Gast and pSom may still display some cross-repression. This is a real outcome, not a failure: it **passes Gate 2** (*i.e.*, GO), has no bifurcation for Gate 3 (*i.e.*, NA), and is reported as such — distinct from a NO-GO where the pattern is absent.

2. The three GO/NO-GO gates (the crux)

Gates are **scientific** (is the phenomenon present?), not parameter-quality checks. Parameter tuning is the agent's means to answer them honestly — never something the researcher gates on; judged visually.

- **Gate 1 (01→02):** After Harmony on `sample` and DR, is the malignant structure both **mixed** (sample-integrated) and **continuous** (connected gradient)? NO-GO if it clusters by sample or breaks into disconnected islands — don't force a trajectory (importance level = HIGH).
- **Gate 2 (03→04):** Does the embryonic module pattern appear — a Gast/pSom hub diverging into two arms, Crest high at tips (ideally pSom arm)? The **non-branching case** (Gast/pSom overlap, Crest an inverse gradient; §1.1) still **passes** (*i.e.*, GO) but has no bifurcation for Gate 3. **Flag** (pass anyway) if Crest enriches the **Gast** arm — an inversion.
- **Gate 3 (04→05, BRANCHING ONLY):** Do tree nodes show **mutual Gast↔pSom negative correlation** (competitive cross-repression)? Crest not tested. No anti-correlation → NO-GO on step 05.

The EMT-TF expression step is unconditional — always produced as the final notebook, outside the gate chain (numbered 06 only when all gates pass; renumbered after an earlier NO-GO — see §1.6).

Linchpin: identify the HUB (*e.g.*, Puram's "Mixed") — the Gast/pSom-overlap convergence population that 04–05 run on and Gate 3 tests. Identify by **role** (module-convergence zone at the trajectory branch point/root), not by name, and confirm it's the topological convergence point.

3. The 6-step protocol (source of truth: Puram2017 notebooks)

`.../CCCA/Puram2017_Head-and-Neck/{01..06}_*.ipynb` (READ-ONLY). Parameter values in reference notebooks are **illustrative, not defaults to copy** — and the **in-notebook comments are illustrative of the analytical judgment calls** the researcher made: the reasoning to emulate rather than transcribe. I/O per step in §1.6.

1. **01_preprocessing**: Load malignant cells → drop under-populated samples (per-dataset min-cell threshold) → filter genes → overdispersed selection → PCA → Harmony on `sample` → neighbors → UMAP → Leiden → drop detached clusters → re-cluster → annotate subtypes. **Annotation always uses both data-derived marker expression and the original publication's reported subtypes/markers** (cited in the log) — never one alone.
2. **02_manifold**: ScFates PPT tree (branching) / curve (non-branching) → biologically-motivated root → pseudotime → rename milestones.
3. **03_signatures**: `score_genes` for Gast/pSom/Crest with the union control pool → signature UMAPs.
4. **04_correlation**: Gast XOR pSom genes, hub cells → pseudotime-windowed module Pearson → correlation UMAP.
5. **05_modules**: Gast OR pSom genes, hub cells → per-gene module correlation → mutual-correlation scatter (reports r).
6. **06_expression**: EMT-TF expression overlay; most informative TFs for that dataset.

Per-dataset notebooks are numbered **sequentially** — the expression step is always the *last* notebook, so its number depends on how far the analysis ran (§1.6).

4. Data & code assets (source of truth)

- **Loader** `.../mesencontext/scripts/load.py` — `load_CCCA_adata(datadir)` reads the matrix/gene/cell files in `datadir`, auto-detects normalization from the matrix filename (TPM/normalized pass through; counts normalized), always `log1p`, sets `celltype` (default 'Malignant'). **Point it at a bottom-level (leaf) dataset directory**, as `merge.py` does — it does not walk nested subdirs. `merge.py` also defines the folder-naming convention mapping summary rows → on-disk folders.
- **Scope list** `.../mesencontext/claude/CCCA_summary.csv` — TAB-sep; `Title, Category, Disease, Technology, Samples, Cells`; one row per in-scope dataset. **Authoritative.**
- **Input data** `.../data/unzip/CCCA/` — one folder per dataset (superset of scope). Layout varies (some datasets nest by technology; expression is a mix of counts and TPM). Resolve each dataset to its leaf matrix directory before loading; don't hardcode format counts.
- **Gene sets** `.../data/features/biomart/` — the four CSVs above.
- **Environment**: The existing `scFates` conda env (§1.5). Runable by setting `R_HOME` path in bash (per notebooks).

5. Directory access & environment

Grant **narrowly, one at a time**; grants apply to **new kernels only** — restart after each.

Directory	Access	Role
`.../data/unzip/CCCA/`	**ro**	input dataset folders
`.../mesencontext/scripts/`	**ro**	`load.py`, `merge.py`
`.../mesencontext/Puram2017_Head-and-Neck/`	**ro**	ground-truth reference notebooks 01–06
`.../data/features/biomart/`	**ro**	gene sets
`.../mesencontext/claude/`	**rw**	notebooks, summary-CSV (copy), cumulative log
`.../data/cancer/vertebrate/CCCA/claude/`	**rw**	output h5ads
`.../figures/mesencontext/claude/`	**rw**	output figures (with `colorbars/`)

Not granted: `EMToolbox` root, `data` root — hence the summary CSV lives in the rw `claude/`.

Environment: Use the existing `scFates` conda env — **no recreation needed**. (Stack: python 3.11, r-base 4.3 with r-mgcv, rpy2, scanpy/scFates/harmonypy/leidenalg/anndata; scFates PPT → rpy2 → r-mgcv, and notebook 01 sets `R_HOME` to the conda prefix.)

6. Deliverables & outputs

Layout: One dataset per subdir, partitioned by `Category`, replicated across three roots (notebooks, h5ads, figures and `colorbars/`): `.../claude/<Category>/<DatasetName>/NN_*.ipynb` (`<DatasetName>` = folder name minus `Data_`). **Relative paths:** The `<Category>/` partition puts notebooks two levels deeper than Puram — adapt the path logic; notebooks must run in place. **Keep rw-directory outputs tidy and as brief as possible** — they live on the researcher's disk; don't accumulate scratch or intermediate files. These deliverables and instructions are **BINDING**.

Step	Reads	h5ad	Figures
preprocessing	raw load	`processed.h5ad`	`samples.pdf`
manifold	`processed.h5ad`	`manifold.h5ad`	`manifold.pdf`
signatures	`manifold.h5ad`	`signatures.h5ad`	`{Gast,pSom,Crest}_signature.pdf`, colorbars
correlation	`signatures.h5ad`	—	`correlation.pdf`, `colorbars/R_colorbar.pdf`
modules	`signatures.h5ad`	—	`modules.pdf`
expression	`processed.h5ad`	—	`expression.pdf`

Stop at a NO-GO: do not create downstream notebooks (not stubbed, not emptied). Notebook numbers are **sequential**, so the unconditional expression step is always the *last* notebook, renumbered to follow whatever preceded it:

- Gate 1 NO-GO → preprocessing (01) and expression (**02**)
- Gate 2 NO-GO → 01–03 and expression (**04**)
- Gate 3 NO-GO → 01–04 and expression (**05**)
- all gates pass → full 01–06 (expression = 06)

Notebooks should contain alternating code and figures — matching the working style of the ground-truth Puram2017 notebooks. These deliverables should reflect the *exact* analysis and results that inform the GO/NO-GO decisions. Any inconsistencies between the notebooks and scientific analyses will **fully invalidate** the results — so treat these notebooks as *critical* assets/deliverables.

Every dataset also produces, regardless of how far it ran:

- `summary.pdf`: A **punchy** one-page executive summary (same register as this doc): dataset in one line (cancer type, platform, malignant-cell count); key parameter choices and rationale (min-cell threshold, root node, geometry); **the gate outcomes with reasoning for each GO/NO-GO**; hub identity and r, where applicable; a one-line verdict on whether the pattern reproduces. **Embeds the step figures** (the produced `.pdf`s) so the decisions are visible at a glance. Format document to match the Puram2017 calibration-step output — this is **BINDING**.
- **one row in** `.../mesencontext/claude/analysis_log.csv` (TAB-sep) — the six `CCCA_summary.csv` columns verbatim (joins 1:1 to the master), then provenance (join key, cell counts through filtering, expression type, key embedding/trajectory params, clusters removed, subtypes and annotation source, geometry) and the gate readout (`gate1/2/3`, Crest-arm flag, hub r, EMT-TFs shown, status, notes).

2 Calibration audit: Puram2017 HNSCC

The autonomous analyst re-ran the six-step protocol on **Puram2017**, the dataset it was hand-built on. Reached step 06, **all three gates GO**. Same data, known answer — so the real question is not *did it pass* but *which passes reflect a decision that will survive contact with data the agent has never seen.*

Verdict. The pattern reproduces: Mixed hub confirmed, cross-repression $\mathbf{r = -0.47}$ (anchor -0.49), hub-and-spoke geometry intact. Cleared to scale, with the graded watch-list below.

Automation criteria: The bar a decision must clear to run unattended

Every dataset-dependent choice must be *automated* for the 60+ dataset run. A choice is safely automated only when it clears four tests; anything short is a liability that a passing gate on the training data (*i.e.*, Puram2017) will hide:

- textbfC1: Derived, not inherited. Computed from the dataset in hand, not copied from the anchor run.
- textbfC2: Shown, not asserted. The derivation is demonstrated in the notebook, not just claimed in a comment (importance = **HIGH**).
- textbfC3: Assumptions stated and met. The rule's structural premises (one hub, one leftover cluster, a clean count gap, a clean bifurcation) are made explicit and hold on the target data. Circularity avoided.
- textbfC4: Portable across scope. Generalizes over the atlas's actual range: cancer types, platforms (10x vs SmartSeq2), and cell counts spanning ~2k to ~90k.

The fixed `Puram2017` skeleton is out of scope here: where the protocol prescribes a value, the agent simply reused it (malignant load 2,539/17 pt, threshold $\geq$100 $\rightarrow$ 2,472, overdispersion 4,804, Harmony/neighbors/UMAP, PPT params, union-pool scoring — down to gene recovery 78/92/103). The criteria apply only to the *tunable* choices below.

Five decisions re-derived

All five are genuinely *derived*, not inherited (C1) — the real achievement. The failures cluster instead on unstated assumptions (C3) and scope (C4), exactly what a training-data calibration cannot expose. Each below: Method, then assessment.

1. **Sample threshold** ($\geq$100 $\rightarrow$ 2,472). *Method:* Sort per-sample malignant cell counts, read the 23$\rightarrow$100 jump as an inflection, keep samples above it. *Assessment:* Depth-invariant in principle, but no gap-finding is actually run (C2) — the "inflection" is just the anchor's $\geq$100 written as a constant; smooth count distributions have no gap, leaving the rule ill-defined (C3).
2. **Island removal** ($-64 \rightarrow$ 2,408, vs reference's hand-listed Leiden -116). *Method:* Build a UMAP radius graph at 6$\times$ the median nearest-neighbour distance, take connected components, keep the largest. *Assessment:* Automated and strictly conservative — strips only disconnected specks — but the 6$\times$ radius is an undeclared constant, and "keep the largest component" assumes noise detaches cleanly (C3): on a dense 60k-cell 10x manifold debris stays attached, or a real second lineage reads as a component and is deleted (C4). Biggest numerical gap (**+52 cells**).
3. **Subtype annotation** (Basal/Classical/Atypical/Mixed). *Method:* Leiden at resolution 0.15, then assign each label to the cluster with the highest mean marker-program score (Basal COL17A1/KRT14/TP63; Classical NRF2 program; Atypical XAGE1 family), greedily, remainder = Mixed. *Assessment:* Robust to Leiden renumbering — no per-dataset re-mapping — but the panels are HNSCC-specific and won't transfer across cancer types (C4), "remainder = Mixed" assumes exactly one leftover cluster (C3), and resolution 0.15 is hardcoded.
4. **Root node** (node 85 vs manual 185; same biology). *Method:* Fit the PPT tree, sum the soft-assignment responsibility of Mixed-hub cells over principal nodes, root at the argmax. *Assessment:* Removes the last human step — the change that makes unattended scale-up viable — but it inherits every upstream error and assumes exactly one convergence point (C3). Most fragile step for unseen datasets.
5. **EMT-TF readout** (ZEB1 and PRRX1 vs hand-picked SNAI1 and ZEB1). *Method:* From an 8-TF candidate panel, score each by arm-specificity (top-arm mean / total arm expression) among expressed factors, take the best TF for two *different* arms. *Assessment:* The procedure generalizes and correctly rejects the high-but-nonspecific SNAI2, but it shows different genes per dataset — so the panel is no longer comparable across the atlas — and presumes a clean bifurcation (C3). Only panel not anchor-matched.

Overfitting: Three choices that fail C1 outright

Three continuous knobs were copied verbatim, rationale asserted but never derived (fail C1 and C2): **PPT (Nodes=200, λ=2000, σ=0.5)** — all scale with cell count, so 200 nodes over 2,408 cells $\neq$ over 60,000; **n_pcs=40 / spread=1.3** — unexamined for larger/sparser sets; **Gate-3 bar r < -0.2** — never tested against a null, and it never bit because -0.47 sits well below it. On the training data an inherited constant passes the gate *because* it was fitted here — the gate certifies nothing about generalization. This is what C1 exists to catch.

Gates & ledger

G1 (mixing/continuity) GO. G2 (module geography) GO — Crest peaks at Basal tip (0.43 vs 0.27 $\approx$ 0.24), Gast highest in Classical, pSom flat. G3 (hub cross-repression) GO — **r = -0.47**,

within 0.02 of the anchor.

Step	Reference	Agent	Divergence
Load / threshold / overdispersion	2,539 · 2,472 · 4,804	2,539 · 2,472 · 4,804	identical
Neighbors / UMAP	n_pcs=40, spread=1.3	same	identical (inherited, fails C1)
Island removal	Leiden 1.5 → 2,356	radius-graph → 2,408	**method · +52**
Subtype annotation	manual ID→label	marker-rule auto	**method (same 4 labels)**
Trajectory (PPT)	Nodes=200 λ=2000 σ=0.5	same	identical (inherited, fails C1)
Root node	185 (manual)	85 (argmax Mixed)	**method (both = hub)**
Gate 3 hub r	−0.49 (n=476)	**−0.47 (n=466)**	**match**
EMT-TF pair	SNAI1/ZEB1 (manual)	ZEB1/PRRX1 (auto)	**readout diverges**

Bottom line. Real automation, not mimicry: all five tunable decisions pass C1. But none was stress-tested on C3/C4, and three hyperparameters fail C1 entirely. Cleared to scale — on the first dataset where the answer is unknown, re-derive the three inherited constants and watch the C3/C4 assumptions (one hub, one leftover cluster, a clean bifurcation, a clean count gap) for the first case that violates them.

3 Evaluation audit: Kürten2021 HNSCC

The analyst ran the six-step protocol on **Kürten2021**, the first dataset whose answer was not known in advance. Reached the expression step, **G1 GO / G2 GO / G3 NA** — the non-branching case (Section 1 §1.1). An expert steered step-by-step and a fresh-context reviewer checked claims. The question is not *did it pass* but *which passes came from the agent's own reasoning* — the gap a fully-autonomous run must close.

Verdict. The pattern reproduces: single embryonic axis, root Gast/pSom-high → terminus Crest-high, EMT-TFs **FOXC1 → SNAI2** — a strict linear subset of Puram. But at nearly every gate the agent's first call was wrong, corrected by the expert or reviewer. Cleared as a *supervised* run; not as evidence of autonomy.

Six forks

1. **Gate 1.** The agent passed G1 on a mixing scalar and a connected-components test; the expert had to redirect attention to the embedding itself, which showed a per-patient archipelago the metrics had scored as integrated. *Rule:* Judge G1 on the regenerated embedding across an integration-strength sweep, with a per-neighborhood metric (iLISI); never on a scalar alone.

2. **Lobe identity.** The integrated manifold resolved into two lobes and the agent moved to fit one curve across both; the expert required each lobe's biological identity first. Only one lobe reproduced the embryonic signature relationships (Gast/pSom co-vary, both oppose Crest); the other was a cycling compartment with Gast confounded by proliferation. *Rule:* A manifold resolves into sub-regions — settle their biology (signatures and cell cycle)

before the G1 GO, and scope the trajectory to the region that reproduces the reference relationships.

3. **Root & topology.** The agent rooted at the Crest tip, treating neural crest as the progenitor state; developmentally Crest is *late*, so the root belongs at the Gast/pSom tip. A non-branching topology is fit with scFates `curve`, not `tree`. *Rule:* Root from the signatures' developmental order; match the scFates method to the topology.

4. **Continuity.** The agent certified the curve as continuous using a metric built from UMAP neighbors, blind by construction to a fold in that same UMAP; the reviewer caught the fold. It arose from re-embedding a linear compartment — an elongated strip a fresh UMAP folds into a U. *Rule:* Do not re-embed a linear compartment, and do not measure continuity with a metric that reads its neighbors off the embedding being tested.

5. **EMT-TF readout.** The canonical Snail/Zeb/Twist panel yielded only a terminus factor; FOXC1, the root-end factor, surfaced only once the panel was widened. *Rule:* Score EMT-TFs on this dataset gated on detectable expression, and include other mesenchymal families such as FOXC and PRRX, not only Snail/Zeb/Twist.

6. **Deliverables.** With G3 = NA, the agent began the branching-only `04_correlation` / `05_modules` notebooks, which assert structure not found. *Rule:* Gate outcomes decide which notebooks exist; an NA or NO-GO means the corresponding deliverables are not produced (§1.6).

Three lessons for an autonomous run

1. **Biology before geometry.** The scientifically decisive forks were the same failure — a geometric result read as biology without the biological check. That check must run and appear in the notebook before the GO: G1 against the regenerated embedding, each subregion's identity before the curve, continuity off an independent embedding. The gate stays binary — the agent proposed "Conditional GO" twice, and it is not an outcome.

2. **Check against the spec, not memory.** The run's deliverable errors were caught by re-reading the binding spec against the notebooks rather than trusting recall — the renumbering rule, the deliverable set, and the summary format were each re-verified against the spec text before finalizing. An autonomous run should re-read the relevant spec section at each gate and before shipping, and check its outputs against the binding text (importance = **HIGH**). An agent must ALWAYS re-read the full spec and context before passing gates — this is now **BINDING**.

3. **Deliverables must be reproducible and consistent.** Each notebook must run end-to-end from raw and reach the reviewed result, and the values it produces must match the summary and log that report them. Where a step is stochastic (Harmony's KMeans is not seed-deterministic), state the tolerance so a benign shift is not read as a changed decision.

Bottom line

The protocol transferred; the judgment did not. The derivation machinery ran correctly on unseen data (threshold by gap-finding, `n_pcs` off the scree, detection-gated signatures), but the decisive calls — which lobe is the manifold, where the trajectory roots, which transcription factors to read — the agent got wrong on first pass, and the expert corrected each by putting

biology ahead of geometry. A fully autonomous run makes those calls without the expert; the three lessons above are what that takes. The verifier is the readiest to automate — the reviewer here was already a fresh-context agent — and is the safeguard that caught the two claims the agent made after asserting the rule against them.

Action items

To safeguard the scientific accuracy of autonomous analyses and maintain FULL compliance with the task definition and calibration/evaluation audits, agents must **ALWAYS** re-read the **FULL** spec before passing a gate, planning an analysis, or writing any deliverable — **THIS IS BINDING**.

- **FAILURE TO REGULARLY REVISIT THE FULL SPEC FILES WILL RESULT IN COMPOUNDING ERRORS AND INCORRECT TASK COMPLETION**

4 Testing audit: Autonomous runs

Each attempt on the **testing** datasets is a fully-autonomous run — the analysis is driven by the spec files alone, with no expert steering. A fresh-context auditor then reviews the run, guided by the expert. One entry per attempt below records the decisive points only; read alongside the three spec files. Datasets are sectioned below, attempts ordered within each.

Chen2020 (NPC, 10x)

Nasopharyngeal carcinoma, 10x, ~3.7k malignant cells — head-and-neck, same tissue family as the Puram/Kürten ground-truth runs.

Attempt 1 · Gate 1 NO-GO

- **Fail: `n_pcs=6` under-powers the manifold.** Chord-elbow over-cut the inherited 40; too few PCs force a patient-segregated read, so the NO-GO is likely an embedding artifact. (Elbow biases low here; the Puram-40 / Kürten-20 precedents are the scale to sanity-check against.)
- **Fail: `summary.pdf` wrong layout.** Verdict-prose and stacked step-PDFs, not the reference gate-ledger → figure-grid one-pager.
- **Held.** Spec re-read at each gate/deliverable; threshold, theta sweep, EMT-TF panel, renumbering all sound.
- **Watch.** Gate call stayed metric-led — embedding read as corroboration, not lead.

Attempt 2 · Gate 1 NO-GO

- **Fail: Integration strength too low.** Settled `theta=2` and swept only `theta 0→3`, dismissing `theta=3` as over-integration without a null test to justify the cutoff. The 10x

precedent (Kürten) reached `theta=10`, so stronger integration was left untested before the NO-GO. The sweep covered `n_pcs` well (attempt-1's issue) but not integration strength.
- **Fail: Figure deliverables diverge from the reference.** `samples.pdf` is a bespoke 3-panel diagnostic (patient UMAP, cell-cycle and `n_pcs`×`theta` sweep) built to argue the audit narrative, not the reference's single sample-colored manifold UMAP; and `summary.pdf` was written to the figures root, not the notebook dir where the reference keeps it.
- **Held.** Both attempt-1 defects fixed: `n_pcs=20` (matches the Kürten/Puram scale) and the `summary.pdf` layout now follows the reference. Gate 1 led with the embedding. Spec re-read/certified throughout.

Attempt 3 · Gate 1 GO / Gate 2 NO-GO / Gate 3 NA

- **Fail: Trajectory mis-specified.** Fit a branching `tree` on a manifold it had itself called non-branching (should be a `curve`), and rooted at a proliferation state rather than the developmental origin the signatures imply — the same root error flagged on Kürten.
- **Fail: Signature scores not smoothed.** Read geography and the Gast/pSom correlation from raw per-cell scores; the Kürten 10x reference graph-smooths the scores (knn-connectivity averaging in 02_manifold) before reading geography, so the raw read is noise-dominated here.
- **Fail: EMT-TF panel degenerate.** The display cutoff was unset, so the color scale is outlier-dominated and washed out; and VIM was shown as an EMT-TF, but vimentin is a mesenchymal marker, not a transcription factor.
- **Fail: Island removal incomplete.** Its metric was degenerate, so detached islands were left in the manifold and Gate 1 was still called continuous.
- **Held.** Deliverables now correct in format and location. Traced the two prior Gate-1 NO-GO calls to an `sc.pp.scale` step before PCA absent from the reference protocol — a correct diagnosis, and why the chain finally ran past Gate 1. Spec re-read/certified throughout.

Attempt 4 · Stopped pre-analysis (no certification)

- **Fail: No certification, immediate drift.** Never certified the binding re-read (it had slipped behind a context fold); drifted into building notebook-execution infrastructure and ran code before any gate. Stopped pre-analysis, then cleaned up correctly.

Attempt 5 · Gate 1 GO / Gate 2 NO-GO / Gate 3 NA

- **Fail: Linear curve fit to a visibly tree-shaped manifold.** Called the clearly Y-shaped manifold "robustly non-branching" on a PPT metric (0 branch nodes across 12 param combos) and fit one curve through it. The inverse of attempt 3's tree-on-a-strip; same cause — topology read from the metric, not the embedding.
- **Fail: Figure deliverables still diverge from the reference layouts, despite the spec re-read.** Re-reading the spec prose did not become a check of the outputs against the binding reference figures themselves.
- **Fail: Gate 2 built toward GO, then reversed to NO-GO on a scalar comparison.** The module geography was legible in the signature embeddings; the call was driven by the scalar comparison instead.
- **Held.** Binding re-read certified at start and at the gate (attempt-4 lesson landed); deliverables in the correct locations; no `sc.pp.scale`; `n_pcs`≈19, theta swept.

Attempt 6 · Gate 1 GO / Gate 2 GO (non-branching) / Gate 3 NA

- **Fail: Folded curve, never caught by any post-hoc visualization.** The committed curve W-folds back through the cloud; `manifold.pdf` draws pseudotime color but not the curve edges, so topology was judged on signature↔pseudotime correlations, blind to the fold (the Kürten failure mode; Section 3). Surfaced only when the expert asked for a curve render in the debrief.
- **Fail: Figure deliverables again diverge; format, content, and tidiness are BINDING.** manifold/samples retitled with milestone and legend labels overlaid on the cloud, not the reference's single panel and external colorbar; an extra `gate2_diagnostics.pdf` left in the figures dir, outside the §1.6 set (scratch, as in attempt 2).
- **Held.** Certified the binding re-read; reached Gate 2 GO (non-branching) matching Kürten; overturned its provisional tree→curve on biology (Gast/pSom co-vary, Crest inverse), not geometry.

Attempt 7 · Gate 1 GO / Gate 2 GO (non-branching) / Gate 3 NA

- **Held: Science lands (first fully-autonomous success).** Correct non-branching read: a fold-free curve rooted at the Gast/pSom-high developmental origin, Gate 2 GO on biology, Gate 3 NA. The Kürten lessons landed — curve not tree, root at origin, fold checked, no re-embedding.
- **Fail: Figure style diverges from the binding reference.** Panels square, not the house `(3.5,1.5)` linear-trajectory aspect; signature UMAPs plot graph-smoothed scores where the reference shows raw (smoothing aids the analysis but misleads as a deliverable); EMT-TF marker size doesn't match the other panels.
- **Fail: `summary.pdf` clips text; tidiness not fully observed.** Verdict and Geometry lines run past the page box, cut off mid-sentence.
- **Watch: One residual island.** A P03-dominated detached lobe survives island removal (39-cell speck stripped, not this); not fatal to the continuous-manifold call.

Bischoff2021 (LUAD, 10x)

Lung adenocarcinoma, 10x, ~13.4k malignant cells — new biology versus the Puram/Kürten/Chen head-and-neck runs.

Attempt 1 · Gate 1 GO (contested) / Gate 2 GO (non-branching) / Gate 3 NA

- **Fail: Pre-Gate-1 error: two-blob manifold passed as continuous.** The manifold splits into two blobs. The agent saw the gap but dismissed it as a UMAP artifact because the kNN graph is one component — deciding a visual gate on a connectivity scalar. The real cause was never settled: either under-integration (a higher θ, rejected for smoothing the gap, may merge the blobs) or a scoping failure (identify each lobe's biology and fit the trajectory only on the lobe reproducing the reference — the Kürten evaluation's lobe-identity fork).
- **Fail: `summary.pdf` rule strikes through the title.** The header underline crosses the title descenders (and the Gate-3 "exclusivity" line); otherwise the deliverable matches the

reference exactly.

- **Held: Deliverables land.** Correct §1.6 set in the correct locations, no scratch, house `(3.5,1.5)` manifold aspect, orientation consistent across UMAPs, raw signature scores shown. Binding re-read certified at start and each gate.

Attempt 2 · Gate 1 NO-GO / Gate 2 NA / Gate 3 NA

- **Fail: Un-integratable call unsupported by its own sweep.** Gate 1 NO-GO ("not continuous") rests on a plateauing diff-sample kNN scalar, yet the sweep's own 16 PC / θ=10 panel shows the gap closing into one continuous mass. The mirror of attempt 1 — scalar over embedding — on the opposite gate.
- **Fail: Sweep mis-centered.** PCs swept only 12/16/20, and <20 is not robust (precedents ran 20/40), so the grid sits below the robust range. The one panel that did converge is its highest PC × highest θ corner, so a properly-centered sweep was never run to confirm it.
- **Fail: `summary.pdf` layout regressed from the reference.** Attempt 1 matched the reference one-pager; this run redesigned it into a bespoke layout (colored gate chips, new title, an extra lobe-fork block).

Attempt 3 · Gate 1 GO / Gate 2 GO (non-branching) / Gate 3 NA

- **Held: Science lands (first Bischoff success).** Correct non-branching call, and it settles the Gate-1 question both prior attempts got wrong: the two density lobes are poles of one continuous manifold, shown on both the embedding and the sweep. Deliverables land — correct §1.6 set, no scratch, reference layout restored.
- **Fail: `summary.pdf` step figures low-resolution.** The six embedded panels were saved at low DPI, so their in-panel text blurs against the sharp vector text of the summary itself.

Sun2021 (HCC, 10x)

Hepatocellular carcinoma, 10x, ~2.3k malignant cells — new biology versus the Puram/Kürten/Chen/Bischoff head-and-neck and lung runs.

Attempt 1 · Gate 1 GO / Gate 2 GO (non-branching) / Gate 3 NA

- **Held: Science lands (first liver success).** Correct non-branching read on new biology: one continuous gastrulation→neural-crest axis rooted at the Gast-high pole. Checked each lobe's biology before certifying (the Kürten fork), kept the full manifold, and self-caught three reviewer findings before re-running.
- **Fail: EMT-TF panel in the wrong plotting style.** A continuous single-gene colormap, not the reference categorical overlay of thresholded expressers on a grey manifold; the overlay branch was coded but bypassed.
- **Fail: `summary.pdf` redesigned, not the reference layout.** Bespoke one-pager, not the certified "GO/NO-GO GATES → STEP FIGURES → Parameter choices" format. Build scratch also left in the notebook dir.
- **BINDING: Verify deliverables against the reference visually.** Compare every deliverable to the reference output directly, not spec prose alone: each panel must match

the reference plotting style, and `summary.pdf` its exact layout. Correct science with divergent panels or layout is not a pass.

5 Full run ledger

Running scorecard — one row per analyzed dataset, reflecting its **last attempt**. Rows are added as datasets are run and finalized (verdicts entered by the expert). Per-run provenance (cell counts, params, subtypes, gate mechanics) lives in `analysis_log.csv`, joined on `JoinKey`.

Title	Category	Disease	JoinKey	Run start	Run end	Expert interaction	Manifold geometry	Gate 1	Gate 2	Gate 3	Hub r	Attempts
Puram et al. 2017	Head and Neck	Head and Neck Squamous Cell Carcinoma	Data_Puram2017_Head-and-Neck	—	—	Teacher	Branching	GO	GO	GO	−0.47	1
Kürten et al. 2021	Head and Neck	Head and Neck Squamous Cell Carcinoma	Data_Kürten2021_Head-and-Neck	2026-07-27 15:08	2026-07-27 22:10	Tutor	Linear	GO	GO	—	—	1
Chen et al. 2020	Head and Neck	Nasopharyngeal Carcinoma	Data_Chen2020_Head-and-Neck	2026-07-29 13:19	2026-07-29 14:47	Reviewer	Linear	GO	GO	—	—	7
Bischoff et al. 2021	Lung	Lung Adenocarcinoma	Data_Bischoff2021_Lung	2026-07-29 22:24	2026-07-29 23:47	Reviewer	Linear	GO	GO	—	—	3
Sun et al. 2021	Liver/Biliary	Hepatocellular Carcinoma	Data_Sun2021_Liver-Biliary	2026-07-30 00:05	2026-07-30 01:23	Reviewer	Linear	GO	GO	—	—	1
Bi et al. 2021 [1]	Kidney	Renal Cell Carcinoma	Data_Bi2021_Kidney	2026-07-30 05:04	2026-07-30 05:48	Judge	Linear	GO	GO	—	—	1
Jansky et al. 2021 [2]	Neuroendocrine	Neuroblastoma	Data_Jansky2021_Neuroendocrine	2026-07-30 13:07	2026-07-30 15:08	Judge	Linear	GO	GO	—	—	1
Tirosh et al. 2016 [3]	Skin	Melanoma	Data_Tirosh2016_Skin	2026-08-05 22:04	2026-08-05 22:49	Judge	Linear	GO	GO	—	—	2
Peng et al. 2019	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Peng2019_Pancreas	2026-08-06 21:22	2026-08-06 22:52	Judge	Linear	GO	GO	—	—	3
Chen et al. 2021	Prostate	Prostate Cancer	Data_Chen2021_Prostate	2026-08-05 20:26	2026-08-05 21:15	Judge	Linear	GO	GO	—	—	2
Lee et al. 2020 [4]	Colorectal	Colorectal Carcinoma	Data_Lee2020_Colorectal	2026-07-30 20:15	2026-07-30 21:51	Judge	Linear	GO	GO	—	—	1
Wang et al. 2019	Brain	Glioblastoma	Data_Wang2019_Brain	2026-08-08 15:20	2026-08-08 17:28	Judge	—	GO	NO-GO	—	—	8
Wu et al. 2021 ‡	Breast	Breast Cancer	Data_Wu2021_Breast	2026-07-31 00:32	2026-07-31 03:03	Judge	Linear	GO	GO	—	—	1
Zhang et al. 2022	Ovarian	High-Grade Serous Ovarian Carcinoma	Data_Zhang2022_Ovarian	2026-07-31 03:15	2026-07-31 13:56	Judge	Linear	GO	GO	—	—	1
Zhou et al. 2020 †	Sarcoma	Osteosarcoma	Data_Zhou2020_Sarcoma	2026-07-31 14:00	2026-07-31 16:14	Judge	Linear	GO	GO	—	—	1
Kim et al. 2020	Lung	Lung Adenocarcinoma	Data_Kim2020_Lung	2026-08-06 21:22	2026-08-06 21:56	Judge	Linear	GO	GO	—	—	3
Steele et al. 2020	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Steele2020_Pancreas	2026-07-31 17:35	2026-07-31 18:33	Judge	Linear	GO	GO	—	—	1

Title	Category	Disease	JoinKey	Run start	Run end	Expert interaction	Manifold geometry	Gate 1	Gate 2	Gate 3	Hub r	Attempts
Song et al. 2022	Prostate	Prostate Cancer	Data_Song2022_Prostate	2026-07-31 18:37	2026-07-31 19:43	Judge	Linear	GO	GO	—	—	1
Ji et al. 2020	Skin	Skin Squamous Cell Carcinoma	Data_Ji2020_Skin	2026-07-31 20:16	2026-07-31 22:01	Judge	Branching	GO	GO	GO	−0.41	10
Lin et al. 2020	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Lin2020_Pancreas	2026-08-01 15:13	2026-08-01 16:27	Judge	Linear	GO	GO	—	—	1
Qian et al. 2020	Lung	Lung Cancer	Data_Qian2020_Lung	2026-08-01 17:17	2026-08-01 18:10	Judge	Linear	GO	GO	—	—	1
Obradovic et al. 2021	Kidney	Renal Cell Carcinoma	Data_Obradovic2021_Kidney	2026-08-08 15:20	2026-08-08 17:51	Judge	Linear	GO	GO	—	—	1
Venteicher et al. 2017	Brain	IDH-Mutant Astrocytoma	Data_Venteicher2017_Brain	2026-08-02 00:23	2026-08-02 01:06	Judge	Linear	GO	GO	—	—	1
Olbrecht et al. 2021	Ovarian	High-Grade Serous Tubo-Ovarian Cancer	Data_Olbrecht2021_Ovarian	2026-08-08 15:20	2026-08-08 19:48	Judge	Linear	GO	GO	—	—	2
Geistlinger et al. 2020	Ovarian	High-Grade Serous Ovarian Carcinoma	Data_Geistlinger2020_Ovarian	2026-08-05 22:04	2026-08-05 22:52	Judge	Linear	GO	GO	—	—	2
Laughney et al. 2020	Lung	Lung Cancer	Data_Laughney2020_Lung	2026-08-02 03:27	2026-08-02 15:58	Judge	Linear	GO	GO	—	—	1
Zhang et al. 2021	Kidney	Renal Cell Carcinoma	Data_Zhang2021_Kidney	2026-08-07 05:53	2026-08-07 07:05	Judge	Linear	GO	GO	—	—	4
Zilionis et al. 2019	Lung	Non-Small Cell Lung Cancer	Data_Zilionis2019_Lung	2026-08-07 23:43	2026-08-08 03:09	Judge	—	GO	NO-GO	—	—	6
Choudhury et al. 2022	Brain	Meningioma	Data_Choudhury2022_Brain	2026-08-02 18:21	2026-08-02 21:46	Judge	—	NO-GO	—	—	—	1
Dong et al. 2020	Prostate	Prostate Cancer	Data_Dong2020_Prostate	2026-08-07 23:43	2026-08-08 02:44	Judge	Linear	GO	GO	—	—	6
Jerby-Arnon et al. 2018 [5]	Skin	Melanoma	Data_Jerby-Arnon2018_Skin	2026-08-02 23:05	2026-08-03 00:08	Judge	Linear	GO	GO	—	—	1
Biermann et al. 2022	Skin	Metastatic Melanoma	Data_Biermann2022_Skin	2026-08-03 00:52	2026-08-03 03:12	Judge	Linear	GO	GO	—	—	1
Maynard et al. 2020	Lung	Lung Cancer	Data_Maynard2020_Lung	2026-08-03 03:33	2026-08-03 12:29	Judge	Linear	GO	GO	—	—	1
Xing et al. 2021	Lung	Lung Cancer	Data_Xing2021_Lung	2026-08-03 12:39	2026-08-03 13:43	Judge	Linear	GO	GO	—	—	1
Chan et al. 2021 [6]	Lung	Lung Cancer	Data_Chan2021_Lung	2026-08-07 05:53	2026-08-07 07:45	Judge	Linear	GO	GO	—	—	4

Title	Category	Disease	JoinKey	Run start	Run end	Expert interaction	Manifold geometry	Gate 1	Gate 2	Gate 3	Hub r	Attempts
Galen et al. 2019	Hematologic	Acute Myeloid Leukemia	Data_Galen2019_Hematologic	2026-08-03 14:21	2026-08-03 15:22	Judge	Linear	GO	GO	—	—	1
Gao et al. 2021	Breast	Breast Cancer	Data_Gao2021_Breast	2026-08-07 05:53	2026-08-07 06:44	Judge	—	NO-GO	—	—	—	4
Hwang et al. 2022	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Hwang2022_Pancreas	2026-08-07 05:53	2026-08-07 10:09	Judge	—	GO	NO-GO	—	—	4
Izar et al. 2020	Ovarian	High-Grade Serous Ovarian Carcinoma	Data_Izar2020_Ovarian	2026-08-03 14:21	2026-08-03 15:03	Judge	Linear	GO	GO	—	—	1
Jerby-Arnon et al. 2021	Sarcoma	Synovial Sarcoma	Data_Jerby-Arnon2021_Sarcoma	2026-08-05 15:19	2026-08-05 16:13	Judge	Linear	GO	GO	—	—	2
Kildisiute et al. 2021	Neuroendocrine	Neuroblastoma	Data_Kildisiute2021_Neuroendocrine	2026-08-05 15:19	2026-08-05 15:52	Judge	—	NO-GO	—	—	—	2
Krishna et al. 2021	Kidney	Renal Cell Carcinoma	Data_Krishna2021_Kidney	2026-08-03 14:21	2026-08-03 15:03	Judge	Linear	GO	GO	—	—	1
Neftel et al. 2019	Brain	Glioblastoma	Data_Neftel2019_Brain	2026-08-03 14:21	2026-08-03 15:25	Judge	Linear	GO	GO	—	—	1
Pelka et al. 2021	Colorectal	Colorectal Cancer	Data_Pelka2021_Colorectal	2026-08-03 14:21	2026-08-03 18:12	Judge	Linear	GO	GO	—	—	1
Olalekan et al. 2021 [7]	Ovarian	Ovarian Cancer	Data_Olalekan2021_Ovarian	2026-08-03 19:57	2026-08-03 20:21	Judge	Linear	GO	GO	—	—	1
Young et al. 2018	Kidney	Kidney Cancer	Data_Young2018_Kidney	2026-08-05 15:19	2026-08-05 15:52	Judge	—	NO-GO	—	—	—	2
He et al. 2021	Prostate	Metastatic Prostate Cancer	Data_He2021_Prostate	2026-08-03 19:57	2026-08-04 13:22	Judge	Linear	GO	GO	—	—	1
Qian et al. 2020	Breast	Breast Cancer	Data_Qian2020_Breast	2026-08-06 21:22	2026-08-06 22:36	Judge	—	NO-GO	—	—	—	3
Paulson et al. 2020	Skin	Merkel Cell Carcinoma	Data_Paulson2020_Skin	2026-08-03 19:57	2026-08-03 20:31	Judge	Linear	GO	GO	—	—	1
Yuan et al. 2018	Brain	High-Grade Glioma	Data_Yuan2018_Brain	2026-08-03 19:57	2026-08-03 20:29	Judge	Linear	GO	GO	—	—	1
Gojo et al. 2020	Brain	Pediatric Ependymoma	Data_Gojo2020_Brain	2026-08-03 19:57	2026-08-04 13:22	Judge	Linear	GO	GO	—	—	1
Raghavan et al. 2021	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Raghavan2021_Pancreas	2026-08-07 18:53	2026-08-07 19:41	Judge	Linear	GO	GO	—	—	5
Cohen et al. 2021	Hematologic	Multiple Myeloma	Data_Cohen2021_Hematologic	2026-08-06 21:22	2026-08-06 23:17	Judge	Linear	GO	GO	—	—	3

Title	Category	Disease	JoinKey	Run start	Run end	Expert interaction	Manifold geometry	Gate 1	Gate 2	Gate 3	Hub r	Attempts
Dong et al. 2020	Neuroendocrine	Neuroblastoma	Data_Dong2020_Neuroendocrine	2026-08-03 19:57	2026-08-03 21:10	Judge	Linear	GO	GO	—	—	1
Couturier et al. 2020[8]	Brain	Glioblastoma	Data_Couturier2020_Brain	2026-08-04 14:02	2026-08-04 15:51	Judge	Linear	GO	GO	—	—	1
Filbin et al. 2018	Brain	Pediatric H3-K27M Mutant Glioma	Data_Filbin2018_Brain	2026-08-04 17:04	2026-08-04 17:12	Judge	Linear	GO	GO	—	—	1
Gaiti et al. 2019	Hematologic	Chronic Lymphocytic Leukemia	Data_Gaiti2019_Hematologic	2026-08-04 14:02	2026-08-04 15:29	Judge	Linear	GO	GO	—	—	1
Liu et al. 2021	Hematologic	Multiple Myeloma	Data_Liu2021_Hematologic	2026-08-05 15:19	2026-08-05 16:20	Judge	—	NO-GO	—	—	—	2
Nath et al. 2021	Ovarian	High-Grade Serous Ovarian Carcinoma	Data_Nath2021_Ovarian	2026-08-04 14:02	2026-08-04 15:36	Judge	Linear	GO	GO	—	—	1
Pal et al. 2021	Breast	Breast Cancer	Data_Pal2021_Breast	2026-08-06 21:22	2026-08-07 04:24	Judge	Linear	GO	GO	—	—	3
Zhang et al. 2019	Ovarian	High-Grade Serous Ovarian Carcinoma	Data_Zhang2019_Ovarian	2026-08-04 14:02	2026-08-04 15:07	Judge	Linear	GO	GO	—	—	1
Li et al. 2017	Colorectal	Colorectal Carcinoma	Data_Li2017_Colorectal	2026-08-04 15:54	2026-08-04 16:20	Judge	Linear	GO	GO	—	—	1
Tirosh et al. 2016	Brain	IDH-Mutant Oligodendroglioma	Data_Tirosh2016_Brain	2026-08-04 15:54	2026-08-04 16:20	Judge	Linear	GO	GO	—	—	1
Bassez et al. 2021	Breast	Breast Cancer	Data_Bassez2021_Breast	2026-08-07 23:43	2026-08-08 03:19	Judge	Linear	GO	GO	—	—	6
Caron et al. 2020	Hematologic	Acute Lymphoblastic Leukemia	Data_Caron2020_Hematologic	2026-08-07 06:57	2026-08-07 10:40	Judge	Linear	GO	GO	—	—	4
Griffiths et al. 2021	Breast	Breast Cancer	Data_Griffiths2021_Breast	2026-08-07 23:43	2026-08-08 04:38	Judge	—	GO	NO-GO	—	—	6
Hovestadt et al. 2019	Brain	Medulloblastoma	Data_Hovestadt2019_Brain	2026-08-04 16:41	2026-08-04 17:16	Judge	Linear	GO	GO	—	—	1
Ledergor et al. 2018	Hematologic	Multiple Myeloma	Data_Ledergor2018_Hematologic	2026-08-05 16:54	2026-08-05 17:49	Judge	Linear	GO	GO	—	—	2
Nam et al. 2019	Hematologic	Myeloproliferative Neoplasms	Data_Nam2019_Hematologic	2026-08-04 16:41	2026-08-04 17:36	Judge	Linear	GO	GO	—	—	1

Title	Category	Disease	JoinKey	Run start	Run end	Expert interaction	Manifold geometry	Gate 1	Gate 2	Gate 3	Hub r	Attempts
Qian et al. 2020	Ovarian	Ovarian Cancer	Data_Qian2020_Ovarian	2026-08-05 16:54	2026-08-05 17:28	Judge	Linear	GO	GO	—	—	2
Rendeiro et al. 2020	Hematologic	Chronic Lymphocytic Leukemia	Data_Rendeiro2020_Hematologic	2026-08-07 23:43	2026-08-08 02:43	Judge	—	NO-GO	—	—	—	6
Roider et al. 2020	Hematologic	B-Cell Lymphoma	Data_Roider2020_Hematologic	2026-08-04 16:41	2026-08-04 17:21	Judge	Linear	GO	GO	—	—	1
Wu et al. 2020	Hematologic	Acute Myeloid Leukemia	Data_Wu2020_Hematologic	2026-08-07 07:47	2026-08-07 10:40	Judge	—	GO	NO-GO	—	—	4
Chung et al. 2017	Breast	Breast Cancer	Data_Chung2017_Breast	2026-08-04 22:59	2026-08-05 01:42	Judge	Linear	GO	GO	—	—	1
Darmanis et al. 2017	Brain	Glioblastoma	Data_Darmanis2017_Brain	2026-08-05 15:47	2026-08-05 15:56	Judge	Linear	GO	GO	—	—	2
Giustacchini et al. 2017	Hematologic	Chronic Myeloid Leukemia	Data_Giustacchini2017_Hematologic	2026-08-05 20:26	2026-08-05 20:49	Judge	Linear	GO	GO	—	—	2
Gulati et al. 2020	Breast	Breast Cancer	Data_Gulati2020_Breast	2026-08-05 22:04	2026-08-05 22:25	Judge	—	GO	NO-GO	—	—	2
Karaayvas et al. 2018	Breast	Breast Cancer	Data_Karaayvas2018_Breast	2026-08-05 15:47	2026-08-05 16:00	Judge	—	GO	NO-GO	—	—	2
Kim et al. 2018	Breast	Triple-Negative Breast Cancer	Data_Kim2018_Breast	2026-08-04 22:23	2026-08-04 23:39	Judge	Linear	GO	GO	—	—	1
Li et al. 2019	Skin	Melanoma	Data_Li2019_Skin	2026-08-07 09:21	2026-08-07 10:40	Judge	—	NO-GO	—	—	—	4
Moncada et al. 2020	Pancreas	Pancreatic Ductal Adenocarcinoma	Data_Moncada2020_Pancreas	2026-08-04 22:55	2026-08-05 01:42	Judge	Linear	GO	GO	—	—	1
Shih et al. 2018	Ovarian	Ovarian Cancer	Data_Shih2018_Ovarian	2026-08-04 22:23	2026-08-04 23:15	Judge	Linear	GO	GO	—	—	1
Song et al. 2019	Lung	Non-Small Cell Lung Cancer	Data_Song2019_Lung	2026-08-04 22:23	2026-08-04 23:04	Judge	Linear	GO	GO	—	—	1
Steen et al. 2021	Hematologic	Diffuse Large B-Cell Lymphoma	Data_Steen2021_Hematologic	2026-08-04 22:23	2026-08-05 01:11	Judge	Linear	GO	GO	—	—	1
Yost et al. 2019	Skin	Skin Carcinoma	Data_Yost2019_Skin	2026-08-04 22:23	2026-08-04 22:54	Judge	Linear	GO	GO	—	—	1
Zhang et al. 2019	Hematologic	Follicular Lymphoma	Data_Zhang2019_Hematologic	2026-08-07 23:43	2026-08-08 02:19	Judge	—	GO	NO-GO	—	—	6

Legend.

- **Expert interaction**: Teacher (calibrated the protocol) · Tutor (steered step-by-step) · Reviewer (autonomous run, audited after the fact) · Judge (outcome adjudicated, no steering).
- **Manifold geometry**: Branching (Gast/pSom hub → pSom arm and Gast arm, cross-repression; Crest a terminal readout at the arm tips) · Linear (collapsed bifurcation, single continuous axis).
- **Gate 1 · Gate 2 · Gate 3**: GO / NO-GO. Gate 3 is branching-only.
- **Hub r**: Gast↔pSom cross-repression Pearson; branching only.
- **Attempts**: Number of runs to reach this finalized row.
- **‡ Wu et al. 2021**: The agent identified an ambiguity in the non-branching definition, which did not say whether Gast/pSom cross-repression is disqualifying. The expert then updated the definition (§1.1–2) to include the cross-repression case — organized cross-repression with a terminal Crest gradient counts as non-branching. Agent-identified, expert-ratified.
- **† Zhou et al. 2020**: Both this run and Zhang 2022 struggled to keep orientation consistent across UMAPs, so the expert made it a binding criterion (Section 4), which this run then met.

Scope corrections. Ireland et al. 2020 (Lung, SCLC) — removed (row, both `analysis_log.csv` rows and all deliverables), dropping the atlas from 87 to 86 rows. It is a cross-species composite (mouse RPM GEM time-course and human sample), so it is not validly in scope.

Prompt Evolution. The autonomous **Judge** runs shared one initiating prompt that set only the regime — scientific instruction stayed in the spec files (except the branching re-run, item 12, which carried scientific steering in the prompt and via debrief). Its substantive edits are numbered below; each carries a superscript on the table row where it first appears (re-run waves overwrite existing rows, so carry none). Operational guardrails (do-not-interrupt, no new packages/environments) are omitted.

1. **Baseline regime**: Named dataset; no-feedback mandate; post-hoc independent-auditor contract. The seed paragraph, carried verbatim thereafter.
2. **Agent-chosen dataset**: Dataset choice delegated to the agent; deliverables first named as the recurring weak spot.
3. **Learn from the log**: Reuse deliverable templates from prior *successful* runs, especially `summary.pdf`.
4. **Science and deliverables**: The flagged weak spot widened from deliverables to include science; this became the canonical template for the single-dataset runs that followed.
5. **Content steer** (one run only): The sole nudge on *what* to analyze: pick a low-pSom dataset. The next run reverted to item 4, unmarked.
6. **Batch delegation / scale-up**: The item-4 template plus one instruction: select 10 datasets and delegate the identical autonomous instructions to 10 subagents, reviewed together. First multi-dataset run.
7. **Delegate self-check**: Item 6 plus one clause: the delegating agent may also check its subagents' science and deliverables accuracy. Second batch.

8. **Full scale-up**: Item 7 unchanged in regime, scope widened from "10 new" to "all remaining in-scope datasets": the final batch, 33 subagents delegated the identical instructions and reviewed together.
9. **Science-fail re-run (`-1`)**: Item-8 regime plus one directive: re-analyze every first-run science failure, hinting most were bad Gate 1/2 GO calls or missed branching.
10. **Science-fail re-run (`-3`)**: Item-9 regime re-applied to the datasets still failing science. Dispatch instructions added strict attention to Gate 1 integration and Gate 2 module geography, cycling-lobe scoping, pseudotime-curve rooting and fold-freeness, and deliverable/biology-visualization fidelity.
11. **Science-fail re-run (`-5`)**: Item-10 regime, dispatch retargeted from pseudotime rooting/fold-freeness to which region the trajectory is scoped on. Several Gate 2 GOs flipped to NO-GO. This exposed a gap in the protocol order: `02_manifold` fits the pseudotime before Gate 2 tests the modules, so the fitted trajectory does not depend on the embryonic modules being present.
12. **Targeted content steer**: For the last run of the bifurcating dataset (Ji), the item-11 regime was narrowed with scientific steering toward a branching outcome (in-prompt and via debrief; see header).